ATOM COUNT CARD

Arm & Hammer Baking Soda ($NaHCO_3$): from a printed label to moles, formula units and atoms

A PRODUCT IDENTITY



Product: Arm & Hammer Baking Soda, net wt 16 oz (1 lb) 454 g
Compound: sodium bicarbonate, $NaHCO_3$
Why it matters: it reacts with acids and releases CO_2 gas, which makes batter rise.
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B BASE QUANTITATIVE DATA

Quantity	Value	Source	Precision
Net weight	454 g	Product label	3 s.f. (limiting)
Purity of compound	100 %	Assumption	Not limiting
Boxes in scale-up	40 boxes	My scenario	Exact count
Atomic masses Na, H, C, O	22.990, 1.008, 12.011, 15.999	CIAAW (2024)	Not limiting
Avogadro constant, N_A	6.022 140 76 × 1023 mol−1	BIPM (2019)	Exact
Formula $NaHCO_3$	subscripts 1, 1, 1, 3	PubChem (n.d.)	Exact counts

Assumption: the contents are 100 % $NaHCO_3$ (the label gives only net weight).

C MOLAR MASS BUILD-UP

Element	Atoms	g/mol each	g/mol	% by mass
Na	1	22.990	22.990	27.367
C	1	1.008	1.008	1.200
H	1	12.011	12.011	14.298
O	3	15.999	47.997	57.135
$NaHCO_3$	**6**		**84.006**	**100.000**

D THE MOLE MAP: ONE 454 G BOX

4.54×10^2 g $NaHCO_3$ → ÷ 84.006 g/mol → **5.40** mol $NaHCO_3$ → × N_A → **3.25×10^{24}** formula units $NaHCO_3$



↓ × 3 O atoms per formula unit

9.76×10^{24} O atoms

E MOLE-CONVERSION CALCULATIONS

1. Molar mass of $NaHCO_3$
1(22.990) + 1(1.008) + 1(12.011) + 3(15.999)
= 22.990 + 1.008 + 12.011 + 47.997 → **84.006 g/mol**

2. Label bridge: package → mass of compound
454 g box × (100 g $NaHCO_3$ / 100 g box)
= 454 g → **4.54×10^2 g $NaHCO_3$**

3. Mass → moles
454 ~~g~~ × (1 mol $NaHCO_3$ / 84.006 ~~g~~)
= 5.4044 mol → **5.40 mol $NaHCO_3$**

4. Moles → formula units
5.4044 ~~mol~~ × (6.022 140 76 × 10^{23} formula units / 1 ~~mol~~)
= 3.2546 × 10^{24} → **3.25×10^{24} formula units $NaHCO_3$**

5. Atoms of one element inside the formula (oxygen)
3.2546 × 10^{24} ~~formula units~~ × (3 O atoms / 1 ~~formula unit~~)
= 9.7638 × 10^{24} → **9.76×10^{24} O atoms**

6. Percent composition + two-route check (oxygen)
%O = [3(15.999) / 84.006] × 100 = 57.135 %
Route A: 454 g × 0.57135 = 259.39 g O
Route B: 5.4044 ~~mol $NaHCO_3$~~ × (3 ~~mol O~~ / 1 ~~mol $NaHCO_3$~~) × (15.999 g O / 1 ~~mol O~~) = 259.39 g O
Both routes → **2.59×10^2 g O** ✓ agree

7. Reverse: % composition → formula (per 100 g)
Na 27.367 ÷ 22.990 = 1.1904 mol
H 1.200 ÷ 1.008 = 1.1905 mol
C 14.298 ÷ 12.011 = 1.1904 mol
O 57.135 ÷ 15.999 = 3.5712 mol
÷ 1.1904 → Na 1.000 : H 1.000 : C 1.000 : O 3.000
→ **$NaHCO_3$** (empirical = molecular, n = 84.006 ÷ 84.006 = 1)

8. Scale-up and reverse
Scale-up: 36 ~~boxes~~ × (5.4044 mol / 1 ~~box~~) = 194.56 mol
→ **1.95×10^2 mol $NaHCO_3$**
Reverse: 1.00 ~~mol~~ × (84.006 g / 1 ~~mol~~) × (1 box / 454 ~~g~~)
= 0.18504 box → **0.185 box**

F SCALE-UP: FROM ONE BOX TO A WHOLE CLASS

1. ONE BOX
454 g of $NaHCO_3$
5.40 mol
3.25 × 10^{24} formula units

2. ONE CLASS
36 boxes (exact count)
1.95×10^2 mol
1.63 × 10^4 g (16.3 kg)

3. PARTICLES, ONE CLASS
36 boxes
1.17×10^{26}
formula units $NaHCO_3$

4. REVERSE
One mole of $NaHCO_3$ is
0.185 box
84.0 g, about 18.5 % of a box

Box counts are exact; 36 boxes (1 per classmate) is my scenario.

G SIGNIFICANT-FIGURE CHECK

Calculator: 3.254 591 × 10^{24} formula units
Reported: **3.25×10^{24} formula units**

- **Limit:** the label's 454 g (3 s.f.).
- **Exact:** N_A, subscripts, box counts.
- Guard digits were kept until the last line of every calculation.

ONE-NUMBER TAKEAWAY

One box holds about 3.25 × 10^{24} formula units, yet one whole mole of baking soda is only 84.0 g, less than a fifth of the box.

H REFERENCES

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